

Yucel Yavuz

AI Data Scientist | Generative AI Practitioner | Machine Learning Specialist

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SUMMARY

Innovative and results-oriented AI Data Scientist with 5+ years of expertise in machine learning, data analytics, and Python-based modeling, complemented by 10+ years of cross-functional leadership experience. Skilled in developing and deploying advanced AI solutions, including LLM-powered agents for media planning, SQL automation, document summarization, and sentiment analysis. Proficient in leveraging Python (Scikit-Learn, XGBoost, SHAP), big data platforms (Databricks, Snowflake, SQL), and BI tools (Tableau, Power BI) to translate complex data into actionable business insights. Adept at applying interpretable models, SHAP analysis, and Generative AI to drive strategic decision-making and measurable impact.

TECHNICAL & SOFT SKILLS

AI, Machine Learning & Analytics: Generative AI • Agentic AI • LangChain • LangGraph • RAG • Snowflake Cortex • Sentiment Analysis • SHAP Explainability • A/B Testing • Supervised & Unsupervised Learning • Logistic Regression • Random Forests • Gradient Boosting • XGBoost • PCA • Cluster Analysis • Time Series Forecasting

Programming Tools & Platforms: Python • PySpark • SQL (Snowflake) • Streamlit • Scikit-Learn • TensorFlow • Keras • H2O.ai • Pandas • NumPy • Databricks • Docker • Git/GitHub • CI/CD

Visualization & BI: Tableau • Power BI • Plotly • Matplotlib • Seaborn • Automated Dashboarding

Leadership: Team Coaching & Mentorship • Cross-Functional Collaboration • Strategic Initiative Leadership • Stakeholder Engagement

PROFESSIONAL EXPERIENCE

Horizon Media – New York, NY

AI Data Scientist | Generative AI | Apr 2023 – Present

- **Architected LLM-driven analytics solutions** (LangChain, Streamlit, Snowflake Cortex) that interpreted demographic and behavioral data, accelerating stakeholder decision-making by **35%** and cutting manual reporting by **40%**.
- **Developed interpretable Databricks AI frameworks using SHAP**, improving model transparency and stakeholder confidence, leading to **20% faster adoption** across analytics workflows.
- **Engineered Databricks–Snowflake pipelines** leveraging **XGBoost, Random Forest, and Gradient Boosting**, achieving **92% accuracy** in LTV and look-alike models, optimizing campaign efficiency by **15%**.
- **Performed segmentation and behavioral analysis** in Databricks (RFM, K-Means, Hierarchical Clustering) identifying high-value audiences that increased retention by **12%**.
- **Designed interactive Tableau and Power BI dashboards** visualizing 1P metrics, cross-sell trends, and behavioral clusters—reducing insight turnaround by **30%** and enabling real-time planning.
- **Delivered ML- and GenAI-driven audience profiling and clustering models** that refined audience definitions and strengthened new business pitches, helping secure **two major client wins** and expand personalization frameworks.
- Built a scalable **Attract–Grow–Retain growth framework** by integrating behavioral, temporal, and transactional insights to uncover key conversion drivers, improving customer lifecycle performance **by ~10%** through personalized strategies.
- **Applied causal inference** techniques (A/B Testing, Propensity Score Matching) to measure true campaign lift and ROI, enabling data-driven allocation decisions that improved targeting performance **by ~12%** and enhanced audience strategy outcomes across the customer lifecycle.

Bowery Farming – Kearny, NJ

Data Scientist | Operations Manager | Sep 2018 – Apr 2023

- Developed predictive yield models comparing **7 ML and 1 DL algorithm**, increasing yield accuracy **by 10%** and improving production planning.
- Built automated **production dashboards**, enhancing productivity **by 25%** and warehouse efficiency **by 20%**.
- Introduced machine learning-driven process optimization that **reduced packaging losses by 20%**.
- Created a Team Lead Development Tool and training program that **promoted 6 employees, reducing turnover to 2%**.
- Created a **great team culture where employees worked efficiently** while ensuring production compliance with company quality, sanitation, and food safety standards [2019 and 2020 SQF (Safe Quality Food) results were 98% and 2021 SQF result was 99% - Excellent].

Teleplan, Mechanicsburg, PA

Sr. Analyst | Supply Chain | Jan 2017 – Sep 2018

- Developed a **supply chain analytics dashboard** improving efficiency by **\$150K** in 8 months and supplier performance to **92.6%**.
- Streamlined forecasting, procurement, and vendor workflows, achieving **99.8%** cycle count accuracy.
- Introduced analytical process improvements that optimized material reuse and cost savings initiatives.

Turkish Army, Ankara/Turkey

Operations and Analytics Manager | Aug 2001-April 2016

- Spearheaded **10+ strategic analytics projects**, saving **\$300K** in resource allocation.
- Designed regression-based forecasting models for personnel and logistics planning.
- Delivered **data-driven reports and dashboards** for executive decision-making across multiple departments.

RECENT PROJECTS

- **SHAP Analysis Workflow – Automated ML Explainability Pipeline on Databricks**

Python, XGBoost, SHAP, PySpark, Databricks AI, Snowflake

Built a Databricks-native end-to-end workflow that trains XGBoost classifiers on 500+ behavioral features from Feature Store, computes SHAP values to rank audience drivers, and leverages Databricks AI (Claude) to auto-generate executive Word reports with strategic insights, business recommendations, and media strategy.

- **Cluster Analysis Workflow – Automated Audience Segmentation Pipeline on Databricks**

Python, PySpark MLlib, Databricks AI, Snowflake

Built a Databricks-native audience segmentation pipeline using PCA + KMeans on 500+ behavioral features from Feature Store, and leveraged Databricks AI to generate executive reports with cluster personas, key insights, and media activation strategies.

- **XGBoost SHAP Model Pipeline – End-to-End ML Scoring on Databricks**

Python, PySpark, XGBoost, SHAP, Databricks Feature Store, Snowflake

Built a Databricks ML pipeline that trains an XGBoost model on 500+ behavioral features, scores the BluSpine population using Pandas UDFs, computes SHAP explainability, and outputs feature importance results to Snowflake.

- **Business Intelligence Chatbot – Text-to-SQL Agent on Databricks**

Python, Streamlit, Databricks AI (ChatDatabricks), LangChain, LangGraph, Plotly

Built a Text-to-SQL conversational BI agent using LangGraph state machine and Databricks-hosted LLMs. Users ask natural language questions; the agent generates and executes SQL, returning results as interactive tables or Plotly charts. Deployed as a Streamlit app on Databricks with AWS.

- **AI Media Planning Agent – Multi-Agent LLM System for Marketing Strategy**

Python, LangGraph, LangChain, OpenAI, Streamlit

Developed a LangGraph-based multi-agent AI system (Supervisor, Business Intelligence Expert, Product Expert, Media Planning Expert, Creative Strategy Expert, Email Writer) to automate end-to-end marketing workflows—turning natural language prompts into data insights, targeted media plans, creative strategy, and personalized email campaigns.

- **Marketing Assistant CoPilot – PDF Insights & Summarization Tool**

Python, Streamlit, LangChain, OpenAI, FAISS

Built a chat-enabled app that allows users to upload PDFs and either summarize documents using customizable templates or query them via natural language. Combined LangChain's RAG architecture with history-aware retrieval, OpenAI LLMs, and FAISS vector search to deliver contextual insights and executive-ready summaries.

EDUCATION

- **U.S. Army War College – Carlisle, PA**

International Fellows Academic Prep Course

- **Turkish National Defense University – Ankara, Turkey**

Ph.D. (Pending), Technology Management

M.S., Technology Management

B.S., Systems Engineering

- **Turkish Army War College – Istanbul, Turkey**

M.A., Organizational Management & Leadership

CERTIFICATIONS

- Generative AI for Data Scientists – *IBM*
- Generative AI Bootcamp – *Business Science*
- Master Machine Learning for Data Science & AI - *CodeBasics*
- Databricks Mosaic AI – *Coursera*
- Certified Management Executive (CME) – *Management & Strategy Institute*